

Software Requirements Specification

AegisOps Bharat

Autonomous AI-Powered Multi-Agency Emergency Disaster Response & GIS Triage Platform

Document Reference	AEGISOPS-SRS-2026-V2
Standard	IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018 Compliant Specification
Version	2.0.0-PROD (Spring Boot 3.3.4 + React 18 Production Edition)
Status	APPROVED / IN OPERATION
Classification	National Critical Infrastructure / Civil Defense Operational Spec

Contents

1	Introduction & Executive Overview	1
1.1	Purpose	1
1.2	Scope of the System	2
1.3	Document Conventions	2
1.4	References	2
2	Overall System Description & High-Level Architecture	2
2.1	Product Perspective	2
2.2	Operational User Roles (RBAC)	4
2.3	Operating Environment	4
3	External Interface Requirements	4
3.1	User Interfaces	4
3.1.1	Full-Screen Immersive War Room (Zero Top Bar)	5
3.1.2	Emergency Request Queue & Data Provenance	5
3.1.3	Mobile Layout Optimization	5
3.2	Hardware & Mobile Device Interfaces	5
3.3	Software & Open Government Telemetry Interfaces	6
3.4	Communications Protocols	6
4	System Features & Functional Requirements	6
4.1	Multi-Channel Report Ingestion	6
4.2	Multilingual Natural Language Processing (NLP) Triage	6
4.3	Multi-Modal Severity Fusion & Dynamic SLA Engine	7
4.4	Automated Tactical Dispatch & Detour Navigation	7
4.5	Hospital Bed Capacity & Mass Casualty Incident (MCI) Surge	7
4.6	Spatiotemporal Disaster Simulation Drill Sandbox	7
4.7	Operator Override & Immutable Audit Governance	8
5	Non-Functional Requirements	8
5.1	Performance & Scalability	8
5.2	Reliability & Fault Tolerance	8
5.3	Security, Privacy & Compliance	8
5.4	Usability & Accessibility	8
6	Data Models, Schemas & Entity Relationships	9
7	Verification & Acceptance Criteria	9
7.1	Backend Java Verification Suite	9
7.2	Frontend React Verification Suite	10

1 Introduction & Executive Overview

1.1 Purpose

This Software Requirements Specification (SRS) establishes the definitive functional, architectural, operational, and non-functional requirements for the **AegisOps Bharat Disaster Coordination Platform**. It serves as the formal contractual and engineering baseline for software architects, backend Java engineers, frontend React developers, control room dispatchers, and state/national civil defense authorities (NDMA, SDMA, DDMA, 112 Command Centers).

1.2 Scope of the System

AegisOps is a mission-critical, enterprise-grade, multi-agency disaster coordination platform designed to drastically compress the critical “golden hour” between hazard occurrence, citizen distress intake, and emergency resource dispatch. The platform integrates:

1. **Multi-Channel Intake:** Citizen Web/PWA, Emergency 112 Voice Helpline, Municipal Helplines (BMC 1916, DDMA 1077, BBMP 1533), and automated messaging bots.
2. **Multilingual AI Triage:** Real-time Natural Language Processing (NLP) across English and Indian regional languages (Hindi, Marathi, Kannada, Tamil, Bengali) extracting casualties, entrapment, and structured equipment needs.
3. **Multi-Modal Severity Fusion:** Mathematical severity scoring (0 — 100) dynamically driving SLA dispatch deadlines.
4. **Geospatial GIS & Open Earth Observation Feeds:** Real-time integration with NASA EONET v3, USGS Seismic Networks, GDACS, IMD Doppler Weather Radar, and OpenStreetMap infrastructure telemetry.
5. **Tactical Fleet & Hospital MCI Surge Management:** Automated nearest-responder dispatch vectors and dynamic trauma bed allocation.
6. **Full Glassmorphic Interface & Zero-Top-Bar Immersive HUD:** High-contrast, theme-adaptive dark/light glassmode optimized for large war room wall screens down to mobile hand-held responder devices.

1.3 Document Conventions

- **MUST / SHALL / REQUIRED:** Mandatory core requirements for production release.
- **SHOULD / RECOMMENDED:** Highly desirable requirements for operational efficiency.
- **MAY / OPTIONAL:** Non-blocking feature extensions.

1.4 References

- IEEE Std 830-1998: *Recommended Practice for Software Requirements Specifications*.
 - ISO/IEC/IEEE 29148:2018: *Systems and Software Engineering — Life Cycle Processes — Requirements Engineering*.
 - National Disaster Management Authority (NDMA) Standard Operating Procedures for Emergency Operations Centers (EOC).
 - Common Alerting Protocol (CAP v1.2) ITU-T Recommendation X.1303.
 - Open Geospatial Consortium (OGC) Standards for Web Map Tile Services & GeoJSON.
 - RFC 6455: *The WebSocket Protocol* & STOMP v1.2 Specification.
-

2 Overall System Description & High-Level Architecture

2.1 Product Perspective

AegisOps operates as a unified Disaster Intelligence command system interfacing between external earth-observation sensor networks, municipal bodies, on-scene emergency responders, and hospital trauma centers.

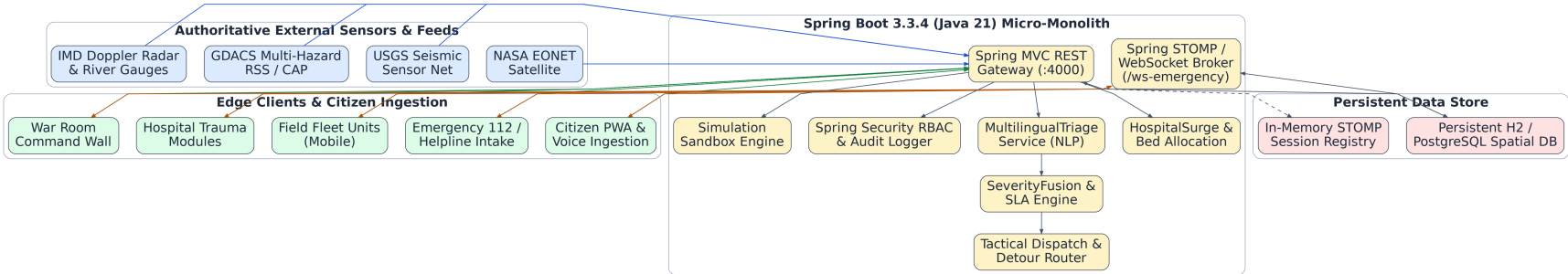


Figure 1: AegisOps high-level architecture: external sensor feeds (blue) and edge clients communicate with the Spring Boot backend over REST (green) and STOMP/WebSocket (orange) channels, backed by persistent spatial storage.

2.2 Operational User Roles (RBAC)

Role ID	Display Name	Access Privileges
CITIZEN	Public Citizen / Guest	Submit emergency reports, track tracking code status, record voice distress audio, view public alerts.
CONTROL_ROOM_OPERATOR	112 EOC Operator	Full read/write triage queue, manual severity overrides, assign rescue fleet units, log audio chimes.
RESCUE_RESPONDER	Tactical Unit Lead	View assigned dispatch orders, update status (EN_ROUTE, ON_SCENE, RESOLVED), report field telemetry.
HOSPITAL_ADMIN	Trauma Bed Director	Update ICU/trauma bed capacities, manage triage tiers (Red/Yellow/Green), trigger Mass Casualty Incident (MCI) surge.
STATE_COMMANDER	State Disaster Chief	Cross-district resource reallocation, state rollup view, approve mutual aid between zones.
NATIONAL_COMMANDER	NDMA Director General	Pan-India National Grid oversight, inject multi-district civil defense simulation drills, broadcast CAP alerts.
SYSTEM_ADMIN	Platform Reliability Eng	Telemetry health monitoring, audit logs inspection, external feeds sync configuration.

2.3 Operating Environment

- **Server Platform:** Linux (Ubuntu 22.04 LTS / Debian 12 / Alpine), containerized via Docker.
- **Java Runtime:** Eclipse Temurin OpenJDK 21.0.4+ LTS.
- **Backend Framework:** Spring Boot 3.3.4 with Spring Data JPA, Spring WebSocket (STOMP), Spring Security.
- **Client Runtime:** Modern evergreen browsers (Chrome 110+, Firefox 115+, Safari 16+, Edge 110+) with WebGL and Service Worker support.
- **Frontend Framework:** React 18.3.1, TypeScript 5.5, Vite 6.4, Leaflet GIS 1.9, Lucide React icons.

3 External Interface Requirements

3.1 User Interfaces

3.1.1 Full-Screen Immersive War Room (Zero Top Bar)

- **Zero-Top-Bar Specification:** The application viewport SHALL eliminate the legacy top command header to maximize the vertical viewing area for GIS map situational awareness and emergency request queues.
- **Sidebar Integration:** All mission controls—Operational Sector/Zone selector (Pan-India, Mumbai, Delhi, Bengaluru, Chennai, Odisha, Kolkata), Mode toggle (LIVE / SIMULATION), Navigation, Emergency Actions, Theme Switcher (Light / Dark), Audio Mute, and Web-Socket live status—SHALL reside within the collapsible frosted glass sidebar (Sidebar.tsx).
- **Glassmode Design System:** All surfaces (sidebar, panels, cards, modals, floating HUD pills) SHALL implement translucent glassmorphism with backdrop-filter: blur(16px) saturate(180%), theme-aware specular borders (rgba(255,255,255,0.08) to rgba(255,255,255,0.85)), and layered ambient box shadows.

3.1.2 Emergency Request Queue & Data Provenance

- **Queue Panel Layout:** The queue SHALL feature an unwrap-proof header displaying the title, live badge counter, and refresh button on row 1, with the triage subtext on row 2.
- **Data Provenance & Source of Information:** Each emergency request card in the queue SHALL explicitly render an authoritative provenance badge:
 - Source channel / agency icon: [Municipal] BMC Disaster Control (Mumbai 1916), [Municipal] DDMA Emergency Helpline (Delhi 1077), [Municipal] BBMP War Room (Bengaluru 1533), [App] Citizen PWA Portal, [Voice] Citizen 112 Helpline, [Bot] WhatsApp Emergency Bot, [Satellite] NASA EONET Satellite, [Sensor] USGS Seismic Feed.
 - Reporter attribution (e.g. Pooja Deshmukh • Verified Caller, Masked Caller Contact).
 - Multilingual NLP extraction model tag (e.g. Hindi NLP, English Multilingual, Kannada Triage, Marathi NLP).
 - Severity color accent: smooth 4px vertical status bar on the left edge with zero border-radius collision artifacts.

3.1.3 Mobile Layout Optimization

- **Responsive Viewport:** On screens with width $\leq 768\text{px}$, the UI SHALL automatically:
 - Conceal desktop sidebar and render a sleek 50px frosted glass .mobile-top-bar with brand logo, hamburger menu, and quick theme toggle.
 - Render a segmented mobile view switcher ([Map View], [Incident Feed] (N), [Split View]) providing full vertical height (calc(100vh - 126px)).
 - Render touch-friendly incident cards with minimum 44px tap targets.
 - Render modals as native mobile bottom-sheets with swipe/scroll friendliness.
 - Anchor a fixed frosted glass .mobile-bottom-nav respecting device safe-area insets (env(safe-area-inset-bottom)).

3.2 Hardware & Mobile Device Interfaces

- **W3C Geolocation API:** Captures high-accuracy GPS coordinates ($\pm 5\text{m}$) for citizen reports.
- **Web Audio API:** Plays multi-frequency auditory alert chimes for critical alerts (≥ 80 risk) unless operator toggles audio mute.
- **Service Worker & Cache API:** Provides progressive web app (PWA) offline intake caching for reports submitted during network blackouts.

3.3 Software & Open Government Telemetry Interfaces

- **NASA EONET v3 REST API:** Ingests active natural events (wildfires, severe storms, flooding) with spatial geometry and hazard categories.
- **USGS Earthquake API:** Ingests seismic events $\geq M3.0$ with magnitude, hypocenter depth, and tsunami advisory status.
- **IMD / Open-Meteo API:** Ingests real-time precipitation (mm/hr), wind velocity (km/h), and atmospheric barometric pressure.
- **GloFAS River Flood Telemetry:** Ingests river discharge rates (m^3/s) with threshold triggers for imminent urban inundation.
- **OpenStreetMap Overpass API:** Harvester for district-level critical infrastructure (hospitals, fire stations, power substations).
- **OSRM Routing Engine:** Generates real-time tactical road corridors and detours impassable flood zones.

3.4 Communications Protocols

- **STOMP 1.2 over SockJS / WebSocket:** Endpoint /ws-emergency handles two-way low-latency telemetry broadcasting:
 - /topic/incidents: Real-time citizen requests and status updates.
 - /topic/resources: Real-time GPS movement and availability of fleet units.
 - /topic/alerts: High-priority NDMA CAP warning broadcasts.
 - /topic/hospitals: Real-time bed occupancy and MCI surge declarations.
- **RESTful HTTPS / JSON:** Endpoints under /api/incidents, /api/resources, /api/hospitals, /api/reports, /api/telemetry, /api/auth.
- **TLS 1.3:** Mandatory transport encryption in production environments.

4 System Features & Functional Requirements

4.1 Multi-Channel Report Ingestion

- **REQ-INGEST-01:** The system SHALL ingest citizen emergency submissions via REST POST /api/reports containing text description, category, latitude, longitude, reporter contact, and optional voice audio payload.
- **REQ-INGEST-02:** The system SHALL assign an immutable unique tracking code formatted as REQ-IND-YYYY-XXXX upon report creation.
- **REQ-INGEST-03:** The system SHALL prevent duplicate incident creation when a report arrives within 1.2 km of an active incident within a 3-hour temporal window.

4.2 Multilingual Natural Language Processing (NLP) Triage

- **REQ-NLP-01:** The system SHALL identify the language of the incoming submission (English, Hindi/Marathi Devanagari, Kannada, Tamil, Bengali).
- **REQ-NLP-02:** The system SHALL parse casualty entities using regularized pattern matchers (e.g. $(\backslash d+)\backslash s^*(\text{injured}|\text{hurt}|\text{ghayal}|\text{casualties})$).
- **REQ-NLP-03:** The system SHALL parse entrapment signals (e.g. $(\backslash d+)\backslash s^*(\text{trapped}|\text{fasc}|\text{stuck}|\text{submerged})$).
- **REQ-NLP-04:** The system SHALL infer structured equipment needs (boats, ALS ambulances, fire engines, extrication jaws, hazmat gear) based on hazard classification and casualty count.

4.3 Multi-Modal Severity Fusion & Dynamic SLA Engine

- **REQ-FUSION-01:** The system SHALL compute an objective composite severity score (0.00 – 100.00) for every incident using the mathematical fusion formula:

$$\text{Score} = \text{clamp}\left(0.40 \cdot W_{\text{base}} + \min(20, 6 \cdot \log_2(N)) + \min(25, 4.5 \cdot C) + \min(25, 4.0 \cdot T) + 15 \cdot D_{\text{cv}}, 10, 100\right)$$

where:

- W_{base} : Baseline hazard weight (Earthquake: 75, Collapse: 70, Fire: 65, Flood: 60, Road Accident: 45, Medical: 50).
- N : Number of corroborating citizen reports merged into the incident cluster.
- C : Count of estimated casualties.
- T : Count of trapped individuals.
- D_{cv} : Normalized computer vision structural damage coefficient ($0.0 \leq D_{\text{cv}} \leq 1.0$).
- **REQ-FUSION-02:** The system SHALL dynamically assign SLA response deadlines based on severity level:
 - **CRITICAL (Score ≥ 80):** 10-minute dispatch SLA target with automated visual & auditory alarm escalation.
 - **HIGH (Score 60 – 79):** 20-minute dispatch SLA target.
 - **MEDIUM (Score 40 – 59):** 35-minute dispatch SLA target.
 - **LOW (Score < 40):** 45-minute dispatch SLA target.
- **REQ-FUSION-03:** When elapsed time exceeds the SLA target, the UI SHALL flag the incident as BREACHED +Xm in high-visibility crimson typography.

4.4 Automated Tactical Dispatch & Detour Navigation

- **REQ-DISP-01:** The system SHALL identify the nearest available emergency response team using the Haversine great-circle distance algorithm:

$$d = 2R \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right)$$

- **REQ-DISP-02:** The system SHALL check if the calculated route intersects an active flood inundation polygon ($\geq 0.5\text{m}$ water level) and compute an alternate detour corridor.
- **REQ-DISP-03:** Upon operator dispatch confirmation, the system SHALL update the team status to DISPATCHED, update the incident status to DISPATCHED, and broadcast the event via WebSocket to all connected operator stations.

4.5 Hospital Bed Capacity & Mass Casualty Incident (MCI) Surge

- **REQ-HOSP-01:** The system SHALL maintain real-time occupancy counts for General Beds, ICU Beds, Ventilators, and Burn Units across municipal hospitals.
- **REQ-HOSP-02:** When incident casualties exceed 10 in a single sector, the system SHALL recommend triggering the **Mass Casualty Incident (MCI) Surge Protocol**, auto-reserving 30% of district emergency beds and initiating inbound patient triage.

4.6 Spatiotemporal Disaster Simulation Drill Sandbox

- **REQ-SIM-01:** The system SHALL support a distinct SIMULATION mode permitting civil defense authorities to inject synthetic multi-hazard disaster scenarios (e.g. Cyclone Tauktae Level-4, Yamuna High-Level Inundation, Metro Subway Collapse) without corrupting live production telemetry.
- **REQ-SIM-02:** Simulation mode SHALL display a persistent high-visibility amber indicator bar and allow single-click rollback to live operational data.

4.7 Operator Override & Immutable Audit Governance

- **REQ-AUDIT-01:** The system SHALL record every administrative modification (severity override, incident reclassification, team reassignment) in an append-only audit trail (AuditLogEntity).
 - **REQ-AUDIT-02:** Every manual override SHALL require an authenticated operator identity, timestamp, previous value, new value, and mandatory operational justification reason.
-

5 Non-Functional Requirements

5.1 Performance & Scalability

- **NFR-PERF-01:** REST API endpoints SHALL exhibit a response latency $\leq 200\text{ms}$ at the 95th percentile under standard operating load (500 req/sec).
- **NFR-PERF-02:** WebSocket STOMP broadcast latency from backend ingestion to frontend screen update SHALL be $\leq 50\text{ms}$.
- **NFR-PERF-03:** The frontend React application SHALL maintain a 60 fps rendering frame rate during map panning, marker clustering, and real-time telemetry updates.

5.2 Reliability & Fault Tolerance

- **NFR-REL-01:** The system SHALL achieve 99.95% operational availability.
- **NFR-REL-02:** The Spring Boot backend SHALL persist data to an ACID-compliant file-backed database (./data/emergency_db) ensuring zero data loss across process restarts.
- **NFR-REL-03:** If external sensor APIs (NASA, USGS, IMD) experience outages or rate limits, the platform SHALL gracefully fallback to historical calibrated baselines without service interruption.

5.3 Security, Privacy & Compliance

- **NFR-SEC-01:** All administrative endpoints SHALL enforce stateless JWT Bearer token authentication and strict Role-Based Access Control (RBAC).
- **NFR-SEC-02:** Passwords SHALL be hashed using salted BCrypt with a minimum cost factor of 12.
- **NFR-SEC-03:** Citizen reporter phone numbers SHALL be masked (e.g. +91-98200-*****) in operator triage queues to prevent PII exposure, decryptable only by authorized dispatchers under audited access.

5.4 Usability & Accessibility

- **NFR-USE-01:** The UI SHALL support both high-contrast Dark Mode and clean frosted Light Mode with zero illegible typography or stark black background bugs.
 - **NFR-USE-02:** The UI SHALL comply with WCAG 2.1 Level AA standards for color contrast and screen reader accessibility tags.
-

6 Data Models, Schemas & Entity Relationships

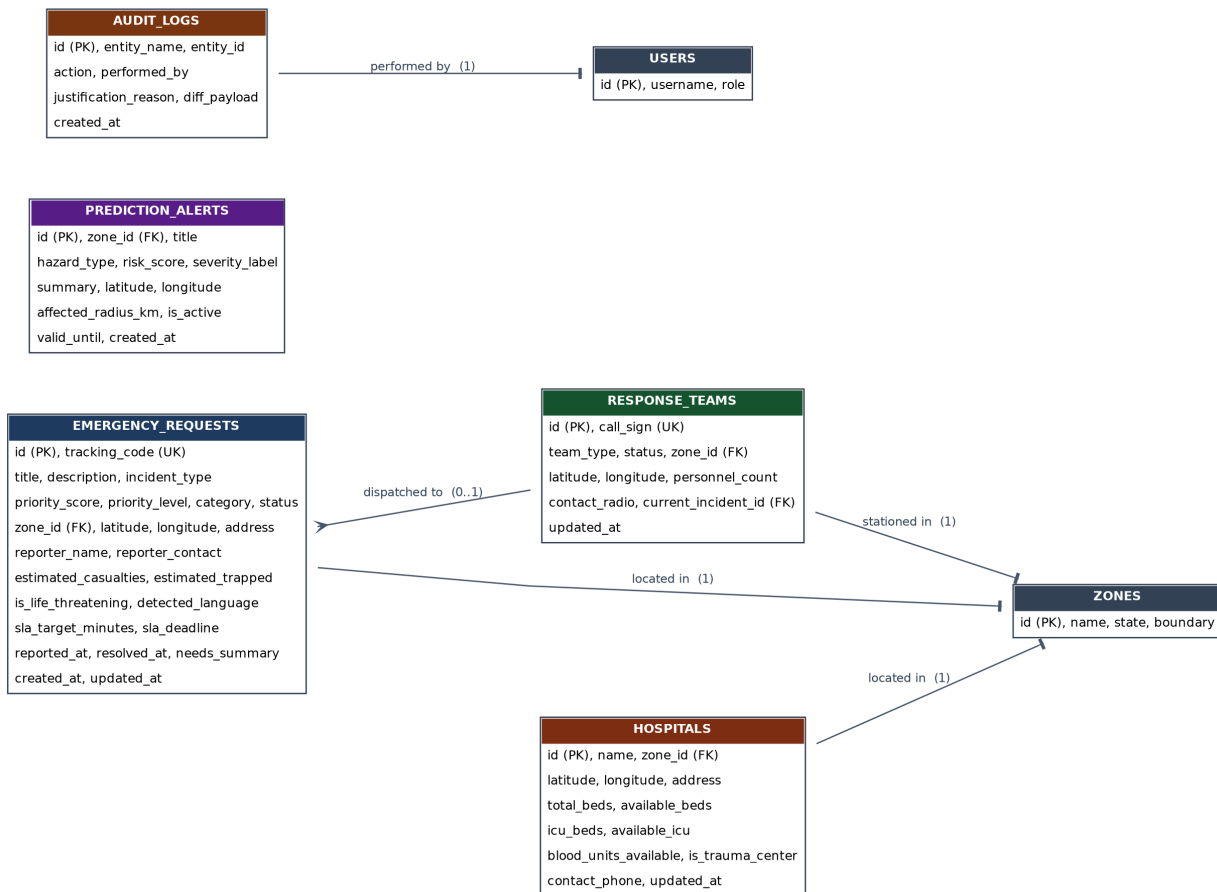


Figure 2: Core relational schema: emergency requests, response teams, hospitals, prediction alerts, and audit logs, all scoped to operational zones.

7 Verification & Acceptance Criteria

7.1 Backend Java Verification Suite

1. **Compilation:** mvn clean package -DskipTests=false passes with 0 compilation errors on OpenJDK 21.
2. **Spring Boot Bootstrap:** Application boots successfully, initializes ./data/emergency_db H2 schema, seeds all 7 disaster zones, 6 fleet response units, and authoritative disaster alerts within ≤ 4.5 seconds.
3. **Endpoint Health:** GET /health returns HTTP 200 with "status": "HEALTHY" and service identification string.
4. **WebSocket STOMP Broker:** Endpoint GET /ws-emergency/info returns HTTP 200 with {"websocket": true}.

7.2 Frontend React Verification Suite

1. **TypeScript Build:** `npm run build (tsc && vite build)` executes cleanly with 0 TypeScript diagnostics and 0 CSS syntax errors.
2. **Immersive HUD Layout:** Top bar is completely removed; GIS map and triage queue render edge-to-edge with 100% viewport coverage.
3. **Source of Info Provenance:** Every emergency request in the queue renders authentic source attribution badge (112 INTAKE, BMC MUNICIPAL, SATELLITE, AI BOT).
4. **Glassmode Theme Switching:** Toggle between Dark Mode and Light Mode verifies zero black artifacts, preserved contrast, and crisp translucent frosted glass.
5. **Mobile Viewport Test:** On 375px — 768px viewports, navigation collapses into sticky glass top bar and bottom nav with zero horizontal scrollbar overflow.

AegisOps Bharat — Standard Operating Documentation — National Civil Defense Systems Architecture.